AIPI 590: Alternative Data
Fall 2026

Dr. Pramod Singh

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Class attendance



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Quick Recap!

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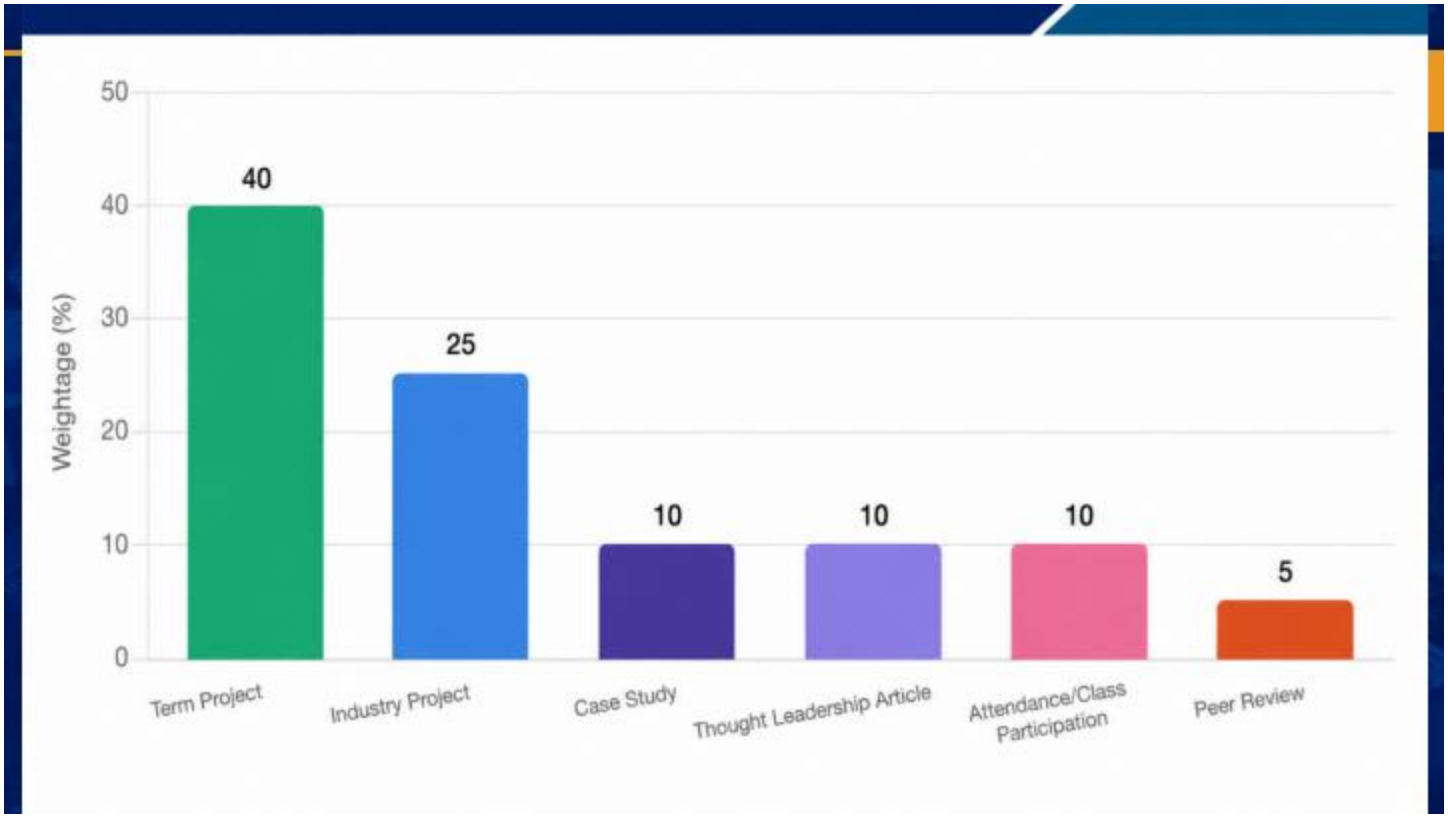
Week	Date	Topic
Week 1	Aug 26, 2026	Foundations of Alternative Data
Week 2	Sep 2, 2026	Types of Alternative Data
Week 3	Sep 9, 2026	The AI Lifecycle in Alternative Data Products
Week 4	Sep 16, 2026	Presentation of Ideas for the Term Project
Week 5	Sep 23, 2026	Investment Research Introduction
Week 6	Sep 30, 2026	Investment Research with Alternative Data – Methods & Pitfalls
Week 7	Oct 7, 2026	Investment Research with Alternative Data – From Models to Strategy
Week 8	Oct 14, 2026	Mid-Semester Presentations
Week 9	Oct 21, 2026	Ethics and Explainability of Alternative Data Products
Week 10	Oct 28, 2026	Case Study Debate & Alternative Data Simulation
Week 11	Nov 4, 2026	Synthetic Data in Alternative Data - Expanding the Edge
Week 12	Nov 11, 2026	Industry Project Presentation
Week 13	Nov 18, 2026	Term Project Presentations

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Course Grading

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Weightage (%)
50
40
30
20
10
0
40
25
10
10
10
5
Term Project
Industry Project
Case Study
Thought Leadership Article
Attendance/Class Participation
Peer Review

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Industry Project

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Your Challenge

Use Thurro's alternative data to build a real, industry-level project

1. **Understand the Problem Statement**
   (Will be shared by today!)
2. **Pull the data**
   the relevant Thurro datasets
3. **Find the signal**
   clean, combine & analyse
4. **Build your Alt Data Product to solve Thurro's challenge**

25% of your grade | Teams will be created by the teaching team

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What better way to understand a company than hearing about it from the co-founder himself?

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Akhilesh Tilotia, Co-Founder @ Thurro

- **20 years** across India's economic transformation including government, sovereign wealth fund, private banking, sell-side, and consulting
- **Co-founder** of **Thurro**, using AI to turn fundamental and alternative data into actionable insights
- Author of ***The Making of India* (2015)** and ***Through the Looking Glass* (2021)**; regular OpEd columnist, Financial Express
- Co-chair, IVCA Climate & Sustainability Council; Distinguished Fellow, Infravision Foundation
- Advisor, Drone Federation of India; co-led IIMA Public Policy Special Interest Group
- **IIM Ahmedabad alum**

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Thurro × Duke

Fall 2026 AltData Class

Akhilesh Tilotia
Co-founder, Thurro
Email • LinkedIn

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Structured Intelligence

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OPENING NOTE

Before we get into the projects

A milestone worth celebrating

Building the curriculum, data access, and infrastructure from scratch and getting it fully operational — a dedicated AltData Class is a real investment. It gives students hands-on access to live, messy, real-world data instead of tidy case studies, and that's not a small thing to pull off. Congratulations to the faculty and course team for getting it off the ground.



Why this partnership excites us

Thurro was built to turn India's messiest, most fragmented alternative data into something structured and queryable. Getting to hand that infrastructure to a new generation of researchers, and seeing what they uncover, is exactly the kind of work we started this company to do.



ABOUT THURRO

Who are we?

Intelligence platform built for research.

8 yrs

Thurro was founded in 2018. It was built and refined for Indian capital markets research.

780+

Datasets across 21 sectors — economy, markets, energy, real estate and more.

470K+

Documents in the research corpus — sell-side notes, credit-rating rationales, DRHPs and filings.

Built in-house

Proprietary AI and data infrastructure, built for real-time processing and updates — not a wrapper on third-party tools.

Client base

Trusted by over 40 sell-side desks, buy-side funds, family offices, and high-frequency trading firms.

India-tested

Proven on one of the world's most complex, fast-moving capital markets — with structured coverage across 21+ sectors and filing types.

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THE CONCEPT

What is alternative data?

Alternative data, or “alt-data,” is information that lives outside financial statements and official government statistics — things like payments, sensor and registration records, satellite imagery, pricing, and text disclosures. It can often be observed before the official numbers come out, or independently of them altogether.

Traditional data

- Quarterly or monthly official releases
- Aggregated at national or state level
- Backward-looking, confirms what happened
- Free or low-cost, well audited

Alternative data

- Daily or weekly signals, near real time
- Granular — district, store, transaction level
- Forward-looking, can lead official prints
- Fragmented, needs cleaning and validation

THE CONCEPT

Where Thurro fits

Thurro exists to close the gap between raw alternative data and a usable research signal. We source, clean, entity-match, and structure India-specific alt-data ourselves, so the heavy data-engineering work is already done by the time analysis begins. For this course, every project connects through one Thurro endpoint, which covers all three capabilities below.



Structured data

The full structured catalogue, ready to query through one SQL endpoint: UPI (India's Unified Payments Interface) transactions, vehicle and property registrations, GST (Goods and Services Tax) collections, grid data, and more.



Financials & market data

Company financials, valuation ratios, shareholding patterns, and mutual fund holdings — all queryable through that same connector.



Semantic documents

Filings, investor calls, and disclosures, all searchable by meaning rather than exact keyword, on that same connector.

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GETTING STARTED

How to access Thurro

So there are no obvious unanswered questions after week 2 — the goal is that any student can pull their first dataset within minutes of class ending.

1

Connect

Add the Thurro connector in Claude (Pro access) or Chat GPT plus. Each team needs at least one member with access — if nobody on a team has it, tell your TA and we'll sort it out.

2

Ask in plain English

No SQL required to get started — ask for a dataset or schema the way you'd ask a colleague. Claude translates the question into a query against Thurro.

3

Get structured answers

Claude returns the data — tables, fields, coverage — ready to pull into Python, R, or a notebook for your own analysis.

ILLUSTRATIVE EXAMPLE

Student: "What's the schema for the UPI transactions dataset?"

Granularity: state, month · Fields: transaction volume, transaction value · History: 3+ years · Update frequency: monthly

Related tables you may also want: vehicle registrations (state, monthly), GST collections (state, monthly)

Five projects, five different edges of alt-data

01

Consumption nowcast



High frequency

02

Rural FMCG signal



Granularity + text

03

Grid backlog → IPP risk



Non-obvious link

04

EV → investment siting



Leading indicator

05

Real estate → cement



Aggregation method

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India high-frequency consumption nowcast

HIGH FREQUENCY

THE QUESTION

Can four real-time payment and logistics signals predict the direction of India's CPI and PFCE numbers before MoSPI, India's Ministry of Statistics and Programme Implementation, publishes them?

> *In plain terms: CPI (Consumer Price Index) tracks how fast prices are rising — inflation. PFCE (Private Final Consumption Expenditure) tracks how much households are spending. Both are official numbers the government publishes with a delay; this project tries to predict their direction early, from faster data.*

THURRO DATASETS USED

- UPI transactions
- Vehicle registrations
- Port & toll logistics
- GST collections

WHY WE PICKED IT

Official consumption numbers lag reality by four to eight weeks for CPI, and by up to two quarters for PFCE. This project asks whether daily and weekly payment and logistics data can close that gap. It's about as pure a test of alt-data's speed advantage as you can build.

PRIMARY DELIVERABLE

Primary: Call CPI's direction 2–4 weeks before MoSPI's release.
Secondary (stretch): Extend the same index to call PFCE's direction too.
Time split: ~75% primary (CPI), ~25% stretch (PFCE).

INDIA It helps RBI (Reserve Bank of India) watchers and corporate planners get ahead of policy and demand signals, weeks before the official numbers land.

UNITED STATES It echoes GDPNow, the Federal Reserve Bank of Atlanta's real-time growth tracker, and the MIDAS method that US macro desks already use to blend data of different frequencies — same idea, applied to a market where the official numbers arrive even later.

MNREGA wage spikes and rural FMCG

GRANULARITY + TEXT

THE QUESTION

Do district-level wage spikes from MNREGA — India's Mahatma Gandhi National Rural Employment Guarantee Act, a rural jobs program — predict which listed FMCG companies will report accelerating rural volume growth, before they say so themselves?

> *In plain terms: MNREGA pays rural households for guaranteed work. When a district's wage payouts spike, those households have more to spend. This project checks whether that extra spending shows up as faster FMCG volume growth before companies mention it on an earnings call.*

WHY WE PICKED IT

Usable rural-income data barely exists, so this signal has to be built from a welfare-scheme dataset, then checked against commentary buried inside earnings-call transcripts. It calls for real entity-matching and text extraction — not just a SQL query.

THURRO DATASETS USED

- MNREGA wages
- Rainfall
- UPI transactions
- Retail distribution
- Earnings-call transcripts

PRIMARY DELIVERABLE

Primary: Flag one FMCG name's rural volume acceleration a full quarter early.
Secondary (stretch): Extend for multiple FMCG names
Time split: 100% primary.

INDIA Distribution and category teams at HUL (Hindustan Unilever), Dabur, Marico, and Emami plan their rural pushes on exactly this kind of income signal.

UNITED STATES It mirrors US retail analysts who track benefit-payment and card-panel spending data to anticipate consumer-staples earnings before companies report them.

Renewable grid-connectivity backlog

NON-OBVIOUS CORRELATION

THE QUESTION

Does a company's position in the queue for a grid connection predict which renewable power producers are most likely to miss their own commissioning guidance?

> *In plain terms: Renewable projects wait in a queue to connect to the grid. The longer that wait drags on, curtailment and REC pricing data start flashing warning signs — often before the company admits it will miss its own start-up date.*

WHY WE PICKED IT

Nothing about a grid-connection application looks like a financial metric. This project project proves an operational bottleneck upstream of the balance sheet quietly predicts predicts guidance risk months before it shows up in results — using curtailment and REC curtailment and REC pricing alongside queue position.

THURRO DATASETS USED

- Grid connectivity queue
- Generation & capacity
- Curtailment
- REC & exchange pricing
- Stock prices

PRIMARY DELIVERABLE

Primary: Flag an IPP likely to miss its own commissioning guidance
Secondary (stretch): Compare with market timing and reaction
Time split: 100% primary.

INDIA Lenders and IPP investors could screen queue exposure as an early-warning flag, ahead of a missed-guidance stock reaction.

UNITED STATES It parallels research from LBNL (Lawrence Berkeley National Laboratory) on interconnection queues, and FERC's (Federal Energy Regulatory Commission) push to reform them — the same bottleneck showing up in US renewable buildout.

EV production and ancillary investment

LEADING INDICATOR

THE QUESTION

Does national growth in EV (electric vehicle) production predict new automotive-sector manufacturing-investment announcements?

> *In plain terms: When more electric vehicles roll off assembly lines nationally, it's often a signal that automakers are about to announce new factories or expansion investments — and this looks at whether that pattern also predicts which states land the money.*

THURRO DATASETS USED

- National production & sales
- State-level registrations
- Industrial investment tracking
- EV pricing & specs

WHY WE PICKED IT

The core, gradeable question is national: does EV production growth predict new investment announcements? The siting question — whether state-level demand explains where capacity lands — is a stretch extension, not a requirement. This keeps the primary deliverable unambiguous while leaving room to test when alt-data actually leads.

PRIMARY DELIVERABLE

Primary: National EV production growth leads investment announcements.
Secondary (stretch): State-level EV demand concentration explains investment gets sited.
Time split: ~80% primary, ~20% stretch — primary stands alone either way.

INDIA State governments courting battery and ancillary plants, and suppliers deciding where to build capacity, are watching this exact tension play out.

UNITED STATES It parallels the gigafactory wave driven by the IRA (Inflation Reduction Act) in Georgia, Michigan, and Tennessee, where incentives, not existing vehicle populations, often decided where plants landed.

Residential launch-to-sales velocity

AGGREGATION METHODOLOGY

THE QUESTION

Does how quickly new homes sell in a city predict the direction of national cement demand, one to two months ahead of the official print?

> *In plain terms: When new homes start selling faster in a city, builders soon need more cement to keep building. This tracks that early signal across cities to call the direction of national cement demand before the official numbers land.*

THURRO DATASETS USED

- Project registrations
- Launch & sales data
- Bank lending rates
- Core Industries Index

WHY WE PICKED IT

The benchmark only exists at a national level, which forces a real methodological choice: equal-weighted versus construction-share-weighted city aggregation. It also demands a discipline most alt-data pitches skip — controlling for lending rates before calling the relationship real.

PRIMARY DELIVERABLE

Primary: Launch-to-sales index calls national cement demand direction 1–2 months early.
Secondary (stretch): Result holds after controlling for lending rates (robustness check).
Time split: ~85% primary, ~15% stretch.

INDIA Cement producers and construction lenders lean on exactly this kind of city-level demand read when making capacity and credit decisions.

UNITED STATES US Census Bureau data on housing starts and permits plays the same leading-indicator role there, with an identical need to separate housing effects from Federal Reserve (Fed) rate cycles.

WHAT WE'VE LEARNED

Best practices for alt-data + AI

The same discipline we ask every project team to apply — and to explicitly test for — in the final build.

Respect the timeline

Only use data that would actually have existed at prediction time. A feature built from next month's number isn't a feature — it's a leak.

Validate before you trust

Every fast signal gets checked against a slower, official benchmark before it's treated as ground truth.

Correlation isn't the finish line

Confounders like rainfall or interest rates get controlled for before we make any leading-indicator claim.

Aggregation is a decision

How granular data rolls up to a national number changes what it predicts. We test more than one way, not just the default.

Structure the unstructured

Some of the best signal lives in filings and call transcripts, not tables. Entity-matching and text extraction are core skills.

Ethics is specific

For every dataset, we check: is this private, could it count as insider information, and was it collected legally? We answer that specifically each time — not with one generic disclaimer for everything.

WORKING TOGETHER

How we'll work together

Office hours: on demand

No fixed weekly slot — email any time with a question and we'll respond. If it's easier to talk it through, we'll set up a quick call.

24-hour response commitment

Expect a reply within 24 hours, most of the time much sooner. If something's going to take longer, we'll tell you so up front.

The LLM is not the answer

Claude and the Thurro connector can pull and shape data fast — but the thinking has to come first. What are you actually trying to test? Structure that before you start querying.

The best projects don't stop at the deadline

Past projects have turned into papers, and a few ideas have grown into small companies after the semester ends. Nothing wrong with stopping at a great grade — but if something clicks, keep going.

DISCLAIMER

Important information

Please read alongside the rest of this proposal.

This presentation is prepared by Adqvest Capital Advisors Pvt Ltd (“Thurro”) solely for the Thurro × Duke AltData Class collaboration. It is for informational and academic purposes only and does not constitute investment advice, a research report, or a solicitation to buy or sell any security. Data, examples, and figures are illustrative of platform capability as of the date shown and are subject to change. Nothing in this document should be relied upon for making investment decisions. Confidential — not for external distribution.

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Thank you

We're looking forward to your comments — and even more, to seeing what the students find.

For office hours, reach out to:

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APPENDIX

Reference material

The detail behind each project, plus a full acronym glossary — for review at your own pace.

Acronym glossary — economy & payments · p.18

Acronym glossary — markets, energy & real estate · p.19

Project 01 — dataset glossary · p.20

Project 02 — dataset glossary · p.21

Project 03 — dataset glossary · p.22

Project 04 — dataset glossary · p.23

Project 05 — dataset glossary · p.24

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APPENDIX — ACRONYM GLOSSARY

Acronyms, in plain English

Economy and payments — what each term means, and the closest US read-across.

Acronym	Full form	What it means	Read across to the US
UPI	Unified Payments Interface	India's real-time, bank-to-bank mobile payment rail — nearly every bank and app plugs into one shared public system.	*Closest everyday feel is Zelle or Venmo, but UPI is an interoperable public utility, not one company's product.*
GST	Goods and Services Tax	India's single nationwide indirect tax on goods and services, replacing a patchwork of state and central taxes.	*The US has no federal VAT/GST — the nearest comparison is the sum of state and local sales taxes, far less unified.*
CPI	Consumer Price Index	Headline measure of retail price inflation, published monthly.	*Functionally the same concept the US BLS publishes monthly, on a similar cadence but a different reporting lag.*
PFCE	Private Final Consumption Expenditure	The household-spending component of GDP, published quarterly.	*Direct analogue is PCE (Personal Consumption Expenditure) — the Fed's own preferred inflation and spending gauge.*
MoSPI	Ministry of Statistics and Programme Implementation	India's central statistical agency; publishes CPI, PFCE, IIP, and GDP.	*Roughly combines what the BLS and BEA do separately in the US.*
RBI	Reserve Bank of India	India's central bank — sets rates, manages the rupee, publishes its own data releases.	*India's Federal Reserve equivalent.*
MNREGA	Mahatma Gandhi National Rural Employment Guarantee Act	A rural jobs-guarantee scheme; wage and employment data doubles as a rural-income proxy.	*No direct modern equivalent — closest historical parallel is a Depression-era public-works program like the WPA.*
IIP	Index of Industrial Production	MoSPI's monthly gauge of factory-sector output, including "core" industries like cement.	*Parallels the Federal Reserve's own Industrial Production Index (the G.17 release).*

Acronyms, in plain English (continued)

Acronym	Full form	What it means	Read across to the US
FMCG	Fast-Moving Consumer Goods	Everyday household and personal-care products with high turnover — soap, snacks, packaged food.	*US industry term for the same category is CPG (Consumer Packaged Goods).*
MNPI	Material Non-Public Information	Undisclosed information that could move a stock price if made public; trading on it is restricted.	*Identical concept and term to what the SEC enforces under US insider-trading law.*
IPP	Independent Power Producer	A company that generates electricity to sell to the grid, rather than a state utility.	*Same term and structure used in US wholesale power markets — merchant IPPs selling into PJM or ERCOT.*
REC	Renewable Energy Certificate	A tradable certificate proving one unit of electricity came from a renewable source, separate from the power itself.	*US states with Renewable Portfolio Standards run comparable REC markets.*
IEX	Indian Energy Exchange	India's power-trading exchange, where RECs and short-term electricity change hands.	*Nearest US parallels are regional wholesale markets like PJM or ERCOT, though those are grid operators as much as exchanges.*
MCLR	Marginal Cost of funds-based Lending Rate	The benchmark Indian banks use to price loans, set relative to their own cost of funds.	*Plays a role similar to the prime rate or a SOFR-linked benchmark in US commercial lending.*
RERA	Real Estate Regulatory Authority	State-level regulator every Indian housing project must register with before it can sell units.	*No single federal equivalent — oversight sits with state real-estate boards, far less centralized.*

PROJECT 01 — DATASET GLOSSARY

What each dataset actually is

Plain-language definitions and platform schema — no prior economics or finance background assumed.

Dataset	What it is	On Thurro	Notes
UPI transactions	India's real-time mobile payments system — similar to Venmo or Zelle. Shows money moved digitally, by state and month.	Monthly · 1,443 rows · Apr 2023–Jun 2026	*State-level*
Vehicle registrations	Every new vehicle registered with the government, by maker, fuel type, and state — a proxy for big-ticket purchases.	Monthly · 1.27M+ rows · Jan 2013–Aug 2026	*Maker / fuel / state level*
Port & toll logistics	Container traffic at India's ports plus FASTag highway toll transactions — both indicate physical goods movement.	Monthly · 1,144 + 53,675 rows · 2019–2026	*Two tables: ports + toll*
GST collections	India's national sales tax, charged on nearly every transaction — a broad, real-time read on economic activity.	Monthly · 1,406 rows · Dec 2024–Jun 2026	*State-level*
Benchmark: MoSPI CPI & PFCE	Official inflation (CPI) and household spending (PFCE) numbers the government publishes with a delay — what this project predicts early.	Monthly (CPI) / Quarterly (PFCE) · 38,508 rows (CPI) · 2013–2026	*PFCE via MOSPI NSA series*

PROJECT 02 — DATASET GLOSSARY

What each dataset actually is

Plain-language definitions and platform schema — no prior economics or finance background assumed.

Dataset	What it is	On Thurro	Notes
MNREGA wages & employment	A government rural-jobs program. District wage and employment data stands in for rural income.	Monthly · 104,916 + 209,414 rows · 2014–2026	*District-level*
Rainfall	Actual vs. normal rainfall, by district — a control since rainfall affects rural income independent of MNREGA.	Daily · 457,873 rows · Dec 2024–Aug 2026	*District-level*
UPI transactions	India's real-time mobile payments system (see Project 01) — used here as a rural/state-level spending signal.	Monthly · 1,443 rows · Apr 2023–Jun 2026	*Same table as Project 01*
Retail distribution	How many stores stock a given brand, by category, state, and city tier — tracks rural expansion.	Weekly · 13M+ rows · Oct 2019–Aug 2026	*Store-location level*
FMCG company financials	Standard quarterly filings for FMCG companies like HUL, Dabur, Marico, and Emami — the benchmark to check against.	Quarterly · standard exchange filings · Ongoing	*P&L, balance sheet, cash flow*
Filings & investor-call corpus	Searchable archive of disclosures and earnings-call transcripts — companies sometimes mention rural growth here.	Continuously updated · not row-counted · Ongoing	*Text corpus, not a structured table*

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PROJECT 03 — DATASET GLOSSARY

What each dataset actually is

Plain-language definitions and platform schema — no prior economics or finance background assumed.

Dataset	What it is	On Thurro	Notes
Grid connectivity applications	Before a solar or wind project can sell power, it applies for a grid connection — a queue that can take years.	Monthly (point-in-time) · 23,742 rows	*Applicant-level*
Renewable generation & capacity	How much renewable capacity exists and how much electricity it's producing, by company, plant, and state.	Monthly · 11,754 rows · Jul 2019–Jul 2026	*Company / station level*
Curtailment	When a plant generates less than it could — usually because the grid can't handle the load, not a lack of sun/wind.	Daily · 2,164 rows · Jan 2025–Aug 2026	*Solar / wind split*
REC & exchange pricing	Tradeable certificates proving a unit of renewable electricity was generated — a signal of plant economics.	Monthly · 162 rows · Jan 2018–Jul 2026	*IEX REC market*
Listed company stock prices	Daily share prices for renewable power companies — used to see if backlog exposure shows up in the market.	Daily · OHLC, all NSE/BSE cos · Ongoing	*Standard pricing table*

PROJECT 04 — DATASET GLOSSARY

What each dataset actually is

Plain-language definitions and platform schema — no prior economics or finance background assumed.

Dataset	What it is	On Thurro	Notes
National production & sales	How many vehicles, by type including electric sub-categories, each manufacturer builds and sells monthly.	Monthly · 14,916 rows · Apr 2023–Jul 2026	*OEM / model level*
State-level registrations	The same vehicle-registration dataset as Project 01, used here to see where EV demand concentrates.	Monthly · 1.27M+ rows · Jan 2013–Aug 2026	*Same table as Project 01*
Industrial investment tracking	Announced manufacturing investment projects — sector, state, cost, status — shows where factories are being built.	Monthly · 3,700 rows	*Project-level*
EV pricing & specifications	Model-level battery capacity and price data, used to identify which models count as electric passenger vehicles.	Weekly · 48,915 rows · Nov 2025–Aug 2026	*Rolling snapshot, not full history*

PROJECT 05 — DATASET GLOSSARY

What each dataset actually is

Plain-language definitions and platform schema — no prior economics or finance background assumed.

Dataset	What it is	On Thurro	Notes
Residential project registrations	Registered housing projects — developer, units, booking status. Indian regulators require this before units sell.	Weekly (state feeds) · KA 1.75M + TG 890K + HR 156K rows	*State RERA feeds + national daily snapshot*
Residential launches & sales	New housing units launched and sold each quarter, by city — a leading indicator of construction activity.	Quarterly · 231 rows · 2018–2026	*7 metro cities (Anarock)*
Bank lending rates (MCLR)	The benchmark rate Indian banks use to price loans — a control since rates affect both real estate and industry.	Monthly · 12,048 rows · Apr 2016–Jul 2026	*Bank-level*
Core Industries Index (cement)	Government index of 8 core infrastructure industries including cement — the benchmark this project predicts early.	Monthly · 1,677 rows · 2023–2026	*Proxy: MOSPI IIP Infra/Construction Goods — not the literal 8-core index*

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Quick Recap (Continued)

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What Do YOU Get Out of This Course?

01

Understand the role of alternative data in AI

Learn how alternative data can fill gaps where traditional datasets are limited.

02

Access Duke's Alternative Data Lab

Explore datasets, tools, and resources available through the lab.

03

Work with real-world data

Gain hands-on experience with high-frequency data provided by industry partners.

04

Build two portfolio-ready projects

Apply what you learn to problems that you can showcase beyond the classroom.

05

Take your ideas further

Past students have continued developing course projects into real products and startup ideas.

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Where alternative data becomes expertise.

What is Alt Data?
The Gap
What We Do

A dedicated graduate course. Research lab. Industry partnerships. Startup incubator. Everything you need to work with alternative data - all in one place, for the first time.

Our Story | Learn More

Get in Touch →

ALT DATA LAB - FALL KICKOFF



Friday, September 4th, 2026

1pm onwards

MEM Lounge (Above Twinnie's)

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Term Project Overview

Experiential learning: the term project is the centerpiece. You'll apply the tools and concepts all semester.

1 Identify
a problem or opportunity

2 Source & validate
unconventional datasets

3 Prototype
analysis & generate insights

4 Present
findings, business-relevant

Format
Individual project or a team of 2

Goal
Turn raw alternative data into clear findings that drive a decision

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Key Milestones & Deliverables

5 %	15 %	20 %
Idea Pitch Sep 16	**Mid-Semester Review** Oct 14	**Final Demo** Nov 18

Deliverables (all decks due before class on the due date)

- **Idea Pitch deck**
- **Mid-Semester deck**
- **Final deck**
- **5-min demo video**
- **GitHub repo**

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Team Assignments

Term project : Please let the teach team know your team's details (or individual) before next class.

Industry project (Thurro) : Team Assignments to be shared before next class!

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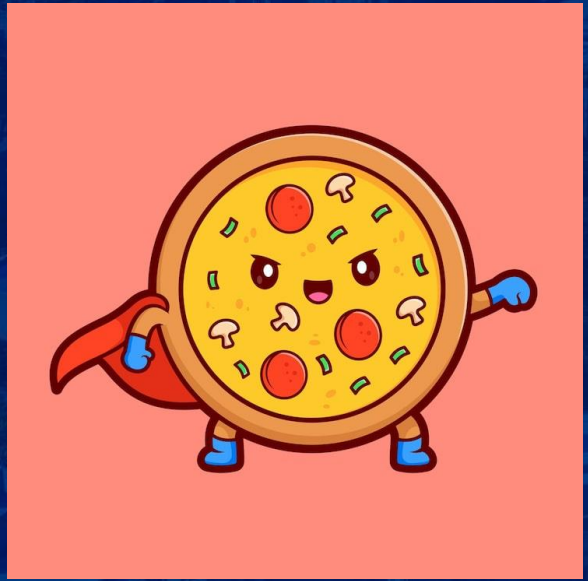


Team Assignments

If you need help forming a team for the term project – reach out to the teach team!

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Out of Class Activity Discussion

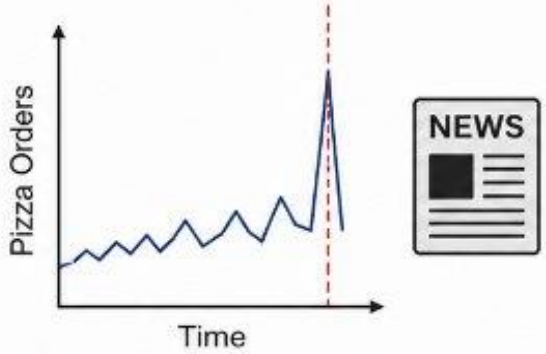
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The “Pizza Meter” Geopolitical Activity



1. Analyze the Case

Read the “Pizza Meter” Theory: Abrupt spikes in late-night pizza orders around government buildings indicating significant national crises. (e.g., historical examples)



2. Propose Validation Methods

Discuss: How would you validate this correlation? (e.g., Historical event matching, statistical significance, temporal/spatial data analysis)



3. Brainstorm Other Signals

Brainstorm: Identify other alternative data signals hinting at major national security crises. (e.g., Satellite imaging of parking lots, localized search volumes, transactional patterns)

What is Alternative Data?

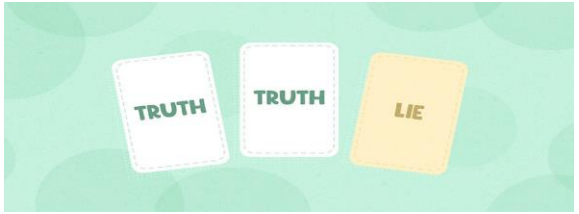
“Alternative data refers to data used by investors to evaluate a company or investment that is not within their traditional data sources.”

Alternatedata.org

“Alternative data is, quite simply, a transformation in how businesses think and make decisions. It turns any source of information into a tool for generating insights and driving business decisions of the future.”

Atlan.com

Warm-Up: Two Truths and a Lie



Three stories about data. One of them never happened!

1

In 2021, Elon Musk tweeted two words: use Signal. A tiny unrelated company called Signal Advance rose thousands of percent in days.

2

In 2018, a fitness app “Strava” published a heatmap of its users’ runs. It traced the perimeter of secret military bases.

3

Walmart found that shoppers buying diapers also buy beer, so it moved the two next to each other and sales jumped.

The Lie: Beer and Diapers



✓ **Signal Advance, 2021: true**

Investors bought a tiny unrelated company because the name was close enough. Mapping data to the right entity is most of the work.

✓ **The Strava heatmap, 2018: true**

Nothing was hacked and nothing leaked. The data was aggregated and anonymous. Aggregated is not the same as anonymous.

✗ **Beer and diapers: folklore**

The most repeated story in data mining, and nobody can produce the study. Always ask where a number came from.

Traditional vs Alternative Data

Traditional

Quarterly earnings

Income statements

EPS

Press releases

Alternative

Google Trends: search interest

App downloads: customer adoption

Social sentiment: brand perception

Employee ratings: culture & health

Job postings: growth vs slowdown

Geolocation: foot traffic

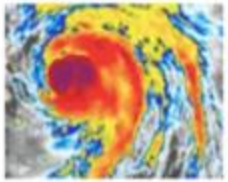
Card transactions: spend ahead of filings



Satellite Data - These are companies that utilize image data from orbiting satellites to do things like measure the number of cars in Walmart parking lots or farm health based on the color of crops.



Web/App/Social Media Data – These are companies which mine social media or use data firehoses from the web/mobile to understand what's happening in the world or how people are interacting with their devices.



Weather Data – These are companies which are developing weather models and utilizing more sensors to get better localized data or improve weather forecasting.



Location/Foot Traffic Data – Companies that use different means to understand where consumers are going by measuring foot traffic via check-ins, video analysis, etc.



Alternative Credit - Companies developing new credit models that utilize sources of alternative data (like mobile usage).



Credit Card Transactions – These are companies that use anonymous aggregate transaction data to understand trends in consumer purchasing habits.



Alternative Data Monetizers/Aggregators – These are companies who pay for access to individual data streams which become more valuable in a bundle, and then sell those packages to investors.



Local Prices – These companies can see what's happening to prices and inflation by aggregating data from ground-level sources.

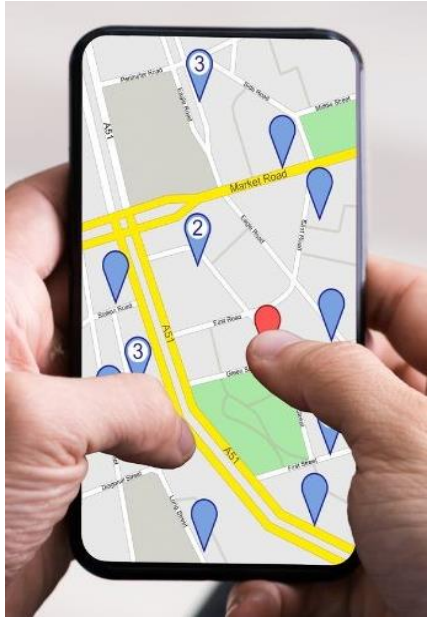
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Geolocation data

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Geolocation: The Everyday Uses



Directions & discovery: Places of interest near me

Social check-ins: Location-tagged posts

Emergency services: 911 location

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Geolocation as Alternative Data

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Geolocation as Alternative Data

What it is

Anonymised, aggregated device-level GPS and Wi-Fi signals collected from mobile apps.

How the Signal Chain Works

1. Device enters store geofence (GPS + Wi-Fi ping)
2. Visit logged if dwell time exceeds 2 minutes
3. Aggregated to store, chain, and category level
4. Normalised against panel size and seasonality
5. Delivered as daily API feed or CSV to fund clients

Geolocation as Alternative Data

Foot Traffic + Card Spend

Links visits to spending

Store-level conversion; brand/ticker performance

Geolocation Trends

Migration hotspots & population shifts

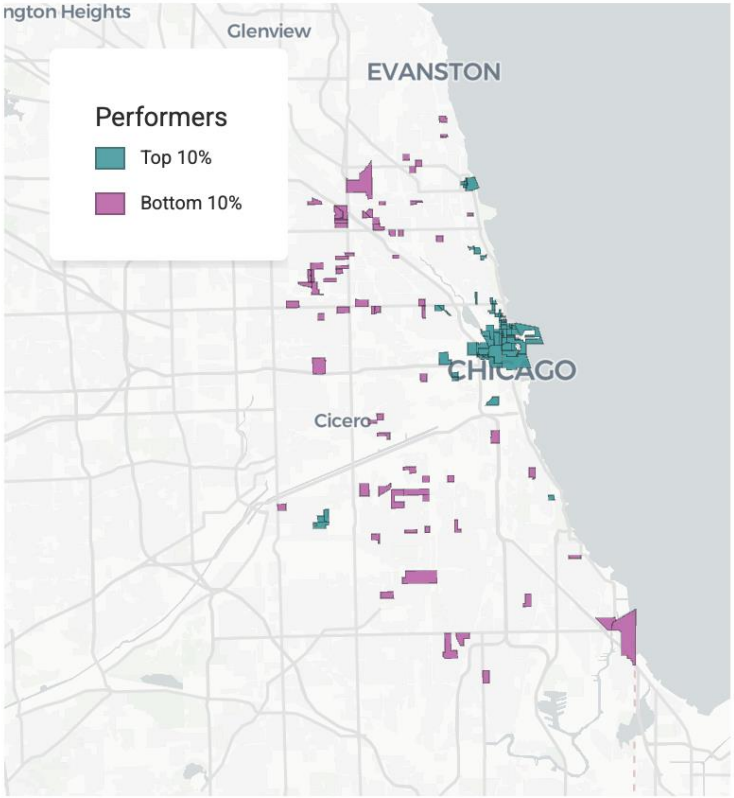
Real estate & housing-demand forecasts

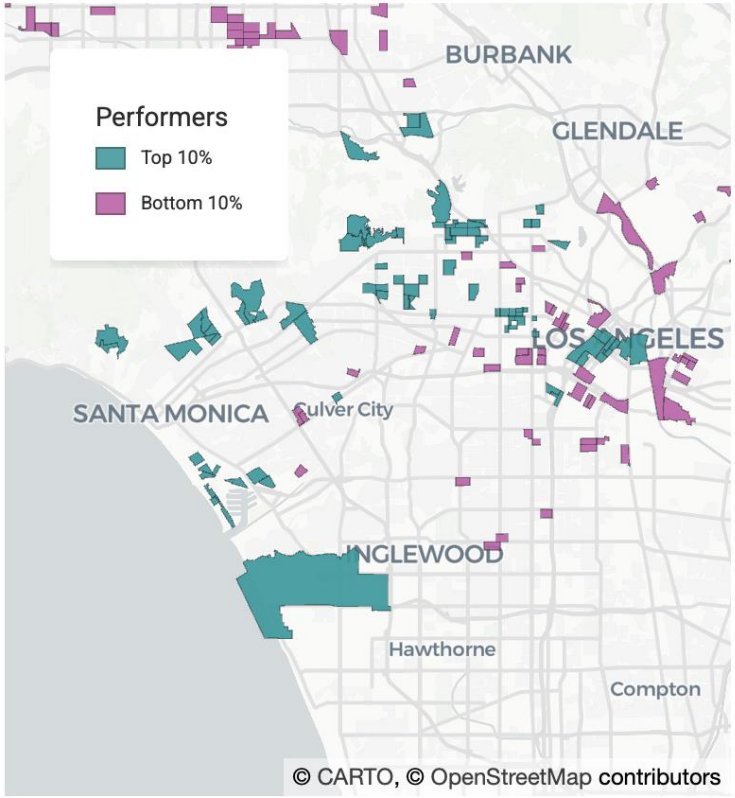
Footfall Density over Time

Best sites for new-store expansion

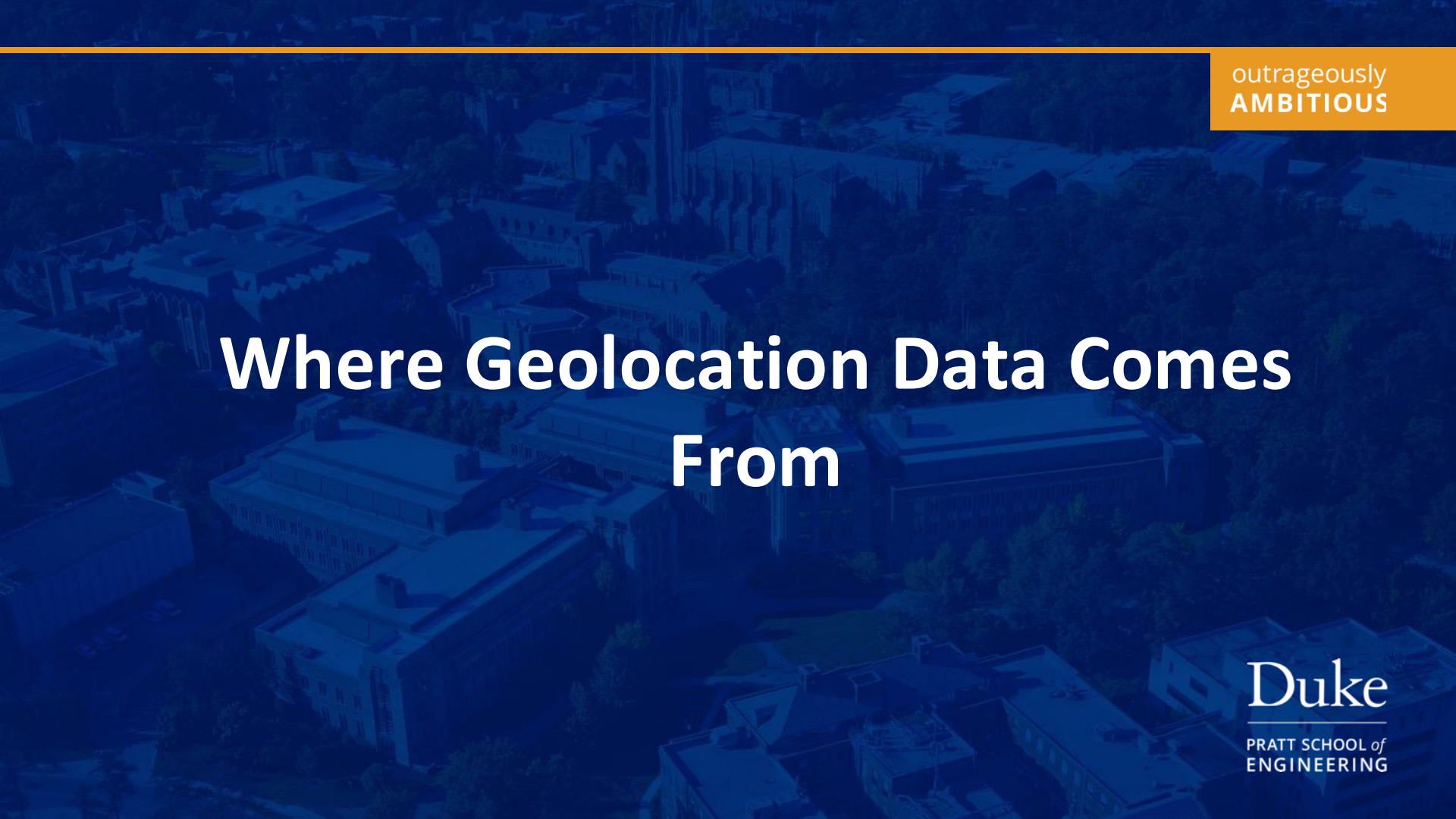
Seasonal & event-driven surges

Using credit card spend alone, can we find the top locations for restaurant sales in Chicago and Los Angeles?





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Where Geolocation Data Comes From
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Where Geolocation Data Comes From

Smartphones

GPS, app tracking, background services

Car Navigation

In-car GPS, telematics data

Wearables

Smartwatches & fitness trackers

Wi-Fi Networks

Hotspot connections & triangulation

Bluetooth Beacons

In-store & proximity tracking

Satellite & IoT

Delivery trucks, scooters, sensors

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Sample Data: Bike Ride Sharing

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rideable_type	started_at	ended_at	start_station_id	end_station_id	start_lat	start_lng	end_lat	end_lng	member_casual	trips_duration	day_of_week
docked_bike	2020-01-21 20:06:59	2020-01-21 20:14:30	239	326	41.9665	-87.6884	41.9671	-87.6674	member	451	3
docked_bike	2020-01-30 14:22:39	2020-01-30 14:26:22	234	318	41.9616	-87.666	41.9542	-87.6644	member	223	5
docked_bike	2020-01-09 19:29:26	2020-01-09 19:32:17	296	117	41.9401	-87.6455	41.9402	-87.653	member	171	5
docked_bike	2020-01-06 16:17:07	2020-01-06 16:25:56	51	24	41.8846	-87.6319	41.8918	-87.6206	member	529	2

Case: Port Congestion (2025)



Signal: Near Intelligence tracked device density at Ningbo-Zhoushan and Shanghai ports. In Q1 2025, worker density dropped 22% below seasonal norms during Lunar New Year. AIS shipping data confirmed vessel queues two weeks later.

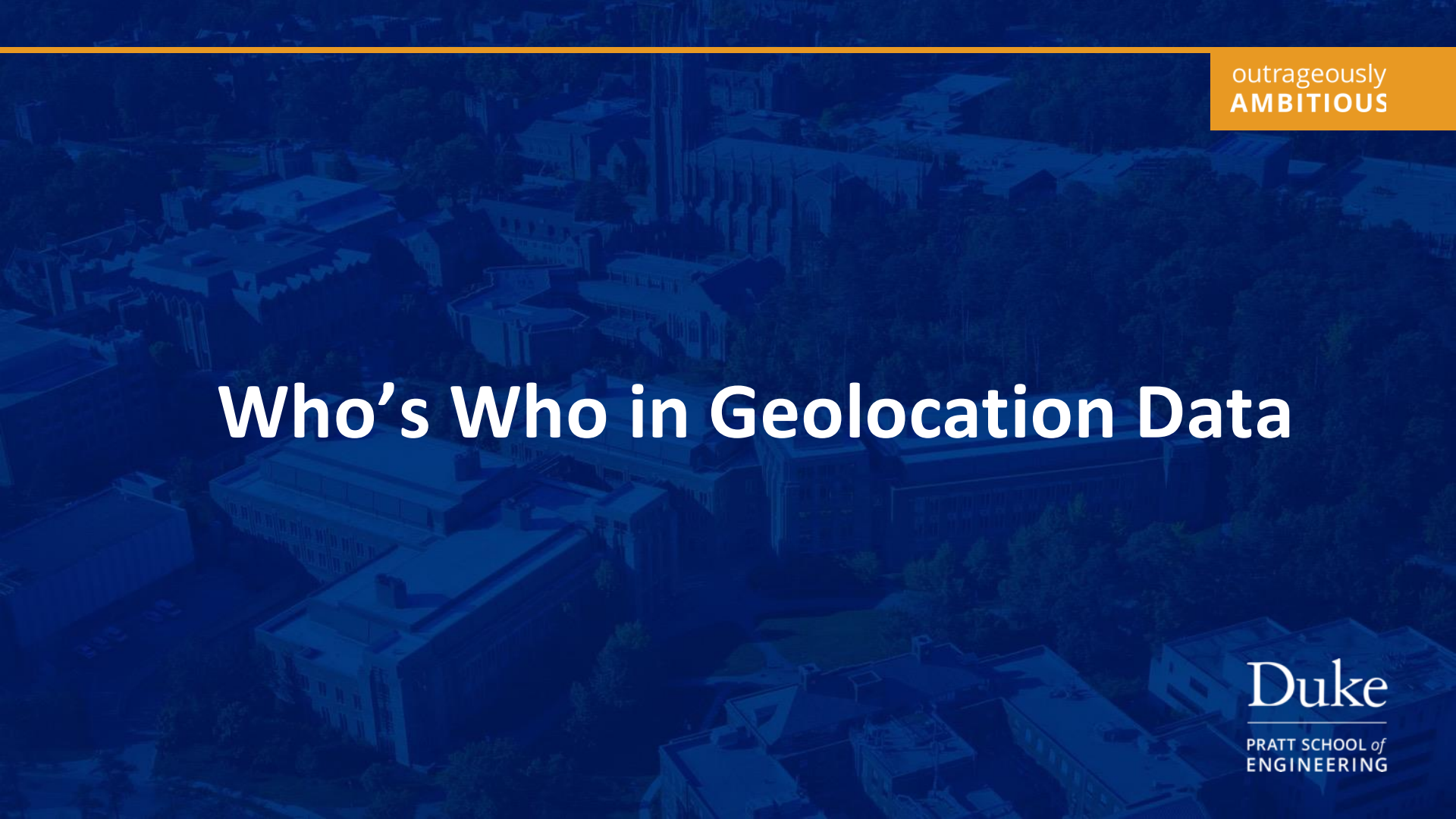
Trade: Short freight-dependent consumer brands (Nike, Gap, Hasbro), long container shipping equities (Maersk, COSCO). Port congestion premium persisted through March 2025.

Case: Target vs Dollar General (2024)



Signal: From Q2 to Q3 2024, Dollar General store visits rose 6.1% YoY while Target visits fell 3.8% in the same ZIP code pairs. The divergence was visible in foot traffic six weeks before both companies reported quarterly same-store sales.

Trade: Funds long Dollar General / short Target ahead of earnings. Dollar General Q3 SSS: +1.3% vs street at -0.5%. Target Q3 SSS: -1.9% vs street at -0.3%. Both moves confirmed the geolocation signal.


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Who's Who in Geolocation Data

Representative vendors in foot-traffic and mobility data (2026)



Placer.ai



Advan Research



Veraset



Unacast



Gravy Analytics



Foursquare



Outlogic



Huq



Dewey



Cuebiq

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Satellite as Alternative Data

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Satellite as Alternative Data

Military Activity: troop movements, base expansions, ship locations

Environmental Monitoring: methane emissions, deforestation, climate impact

Retail Analysis: parking-lot traffic to predict big-retailer sales

Infrastructure & Construction: new projects, urban growth, illegal mining


TARGET


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Where Satellite Data Comes From

Raw Satellite Data

Landsat: long-running Earth observation (NASA / USGS)

Sentinel: EU Copernicus program

Access & Analysis Platforms

Planetary Computer: Microsoft's cloud geospatial access

Google Earth Engine: large-scale geospatial analysis

Sentinel Hub: APIs for Sentinel & Landsat

Case: Counting Oil From Space



How it works: Crude oil storage tanks use floating roofs that sit directly on the oil surface. As inventory rises or falls, the roof moves. The shadow cast between the roof rim and the tank wall changes in proportion to fill level - measurable by satellite using computer vision.

Signal : Satellite inventory data showed crude surplus building in Shandong province storage terminals across six consecutive weeks in Q2 2023.

Trade : Q2 2023 Shandong province crude surplus was visible in satellite data for six consecutive weeks before any official Chinese customs, OPEC, or EIA figure reflected it. Funds short Brent crude futures from the satellite signal captured the 8.3% decline when official data eventually confirmed.

Case: Counting Cars in Parking Lots

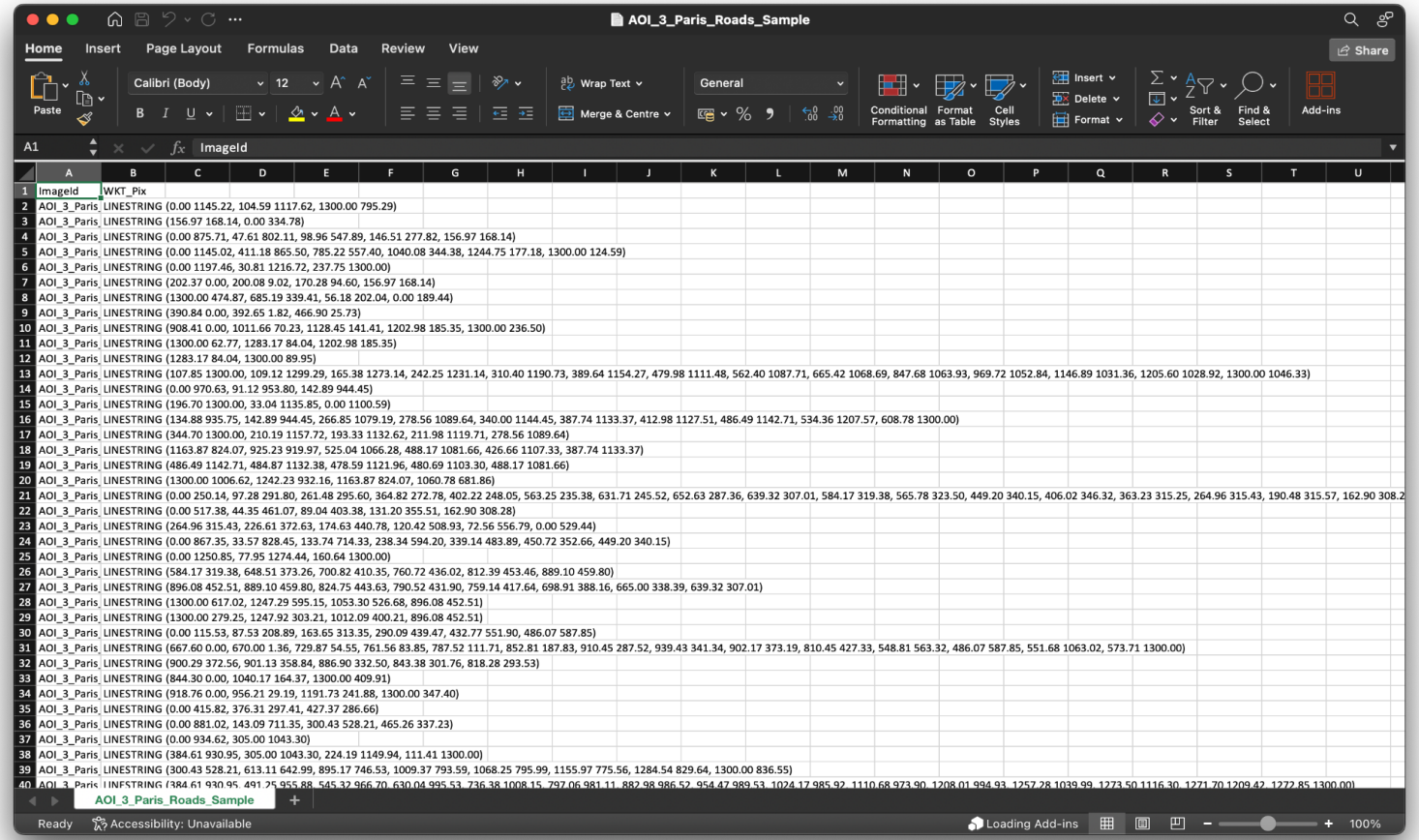


How it works: Planet Labs and Maxar satellites photograph retail store parking lots daily at 3-5 metre resolution. Computer vision models count vehicles and estimate occupancy. Compared to prior-year counts at the same location, vehicle counts become a same-store sales (SSS) proxy. RS Metrics monitors 5,000+ US retail locations. SpaceKnow covers global retail chains.

Signal : RS Metrics satellite counts showed Target parking lots running 12% below prior year while Walmart lots ran 6% above -- for six consecutive weeks before Q3 earnings.

Trade : HORT Target (TGT) / LONG Walmart (WMT). Target Q3 SSS: -1.9% (missed by 1.6pts). Walmart Q3 SSS: +4.9% (beat by 1.1pts). TGT fell 8.1% on earnings day.


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Sample Data: Satellite Road Imagery
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AOI_3_Paris_Roads_Sample

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A1 ImageId

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16	AOI_3_Paris	LINESTRING (134.88 935.75, 142.89 944.45, 266.85 1079.19, 278.56 1089.64, 340.00 1144.45, 387.74 1133.37, 412.98 1127.51, 486.49 1142.71, 534.36 1207.57, 608.78 1300.00)
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18	AOI_3_Paris	LINESTRING (1163.87 824.07, 925.23 919.97, 525.04 1066.28, 488.17 1081.66, 426.66 1107.33, 387.74 1133.37)
19	AOI_3_Paris	LINESTRING (486.49 1142.71, 484.87 1132.38, 478.59 1121.96, 480.69 1103.30, 488.17 1081.66)
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21	AOI_3_Paris	LINESTRING (0.00 250.14, 97.28 291.80, 261.48 295.60, 364.82 272.78, 402.22 248.05, 563.25 235.38, 631.71 245.52, 652.63 287.36, 639.32 307.01, 584.17 319.38, 565.78 323.50, 449.20 340.15, 406.02 346.32, 363.23 315.25, 264.96 315.43, 190.48 315.57, 162.90 308.2
22	AOI_3_Paris	LINESTRING (0.00 517.38, 44.35 461.07, 89.04 403.38, 131.20 355.51, 162.90 308.28)
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29	AOI_3_Paris	LINESTRING (1300.00 279.25, 1247.92 303.21, 1012.09 400.21, 896.08 452.51)
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40	AOI_3_Paris	LINESTRING (384.61 930.95, 491.25 955.88, 545.32 966.70, 630.04 995.53, 736.38 1008.15, 797.06 981.11, 882.98 986.52, 954.47 989.53, 1024.17 985.92, 1110.68 973.90, 1208.01 994.93, 1257.28 1039.99, 1273.50 1116.30, 1271.70 1209.42, 1272.85 1300.00)

AOI_3_Paris_Roads_Sample

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Who’s Who in Satellite Data
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Who's Who in Satellite Data

Representative vendors in satellite and geospatial imagery (2026)



Planet Labs



Maxar (Vantor)



BlackSky



Satellogic



ICEYE



Ursa Space



Kpler



Spire Global



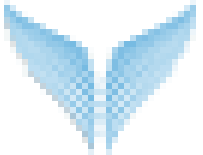
RS Metrics

SPACE KNOW

SpaceKnow



Descartes Labs



Kayrros

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From Data to Discovery

Now let’s see how satellite data is applied in cutting-edge research.

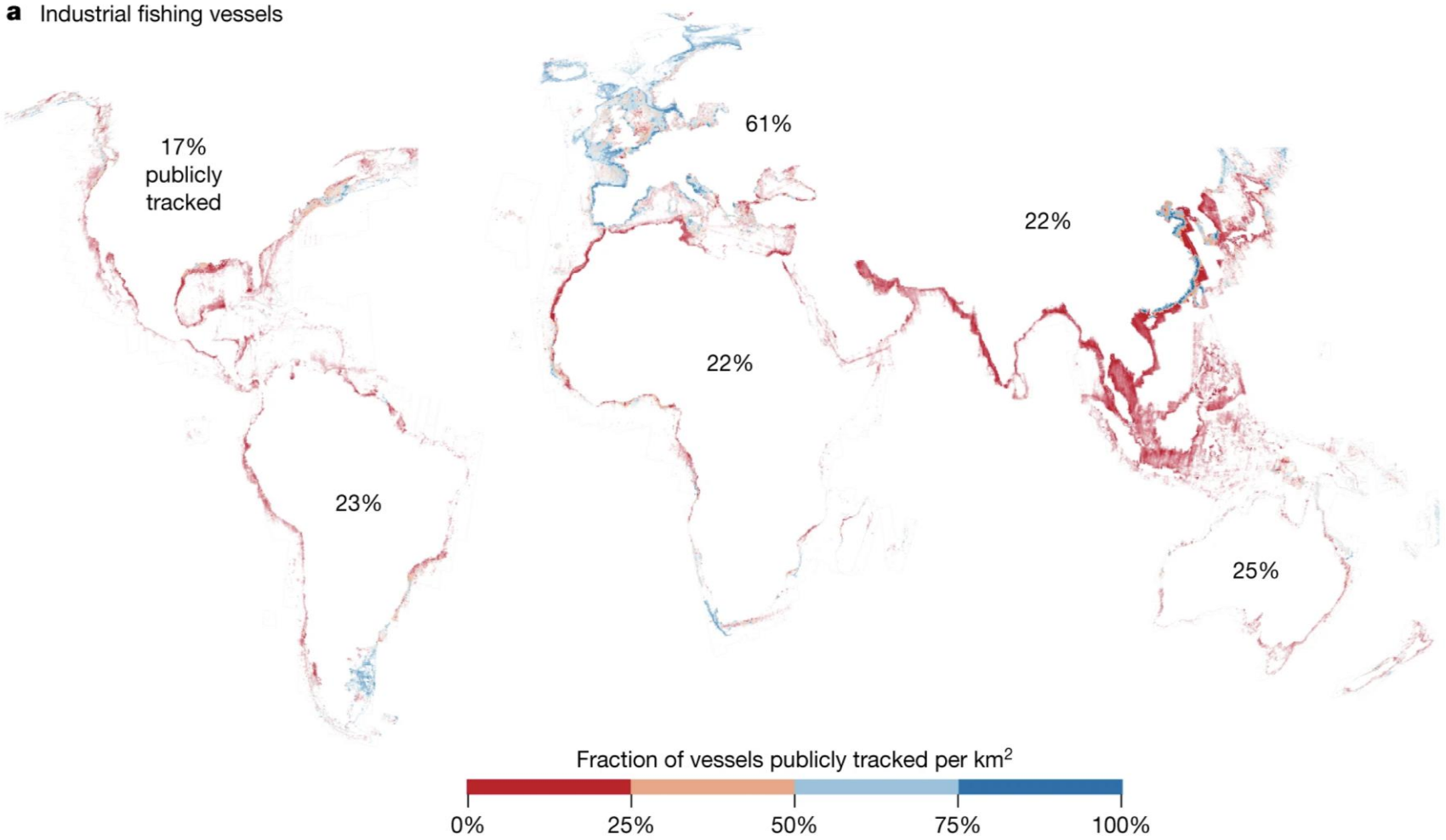
A recent **Nature** study reveals patterns we would otherwise miss.

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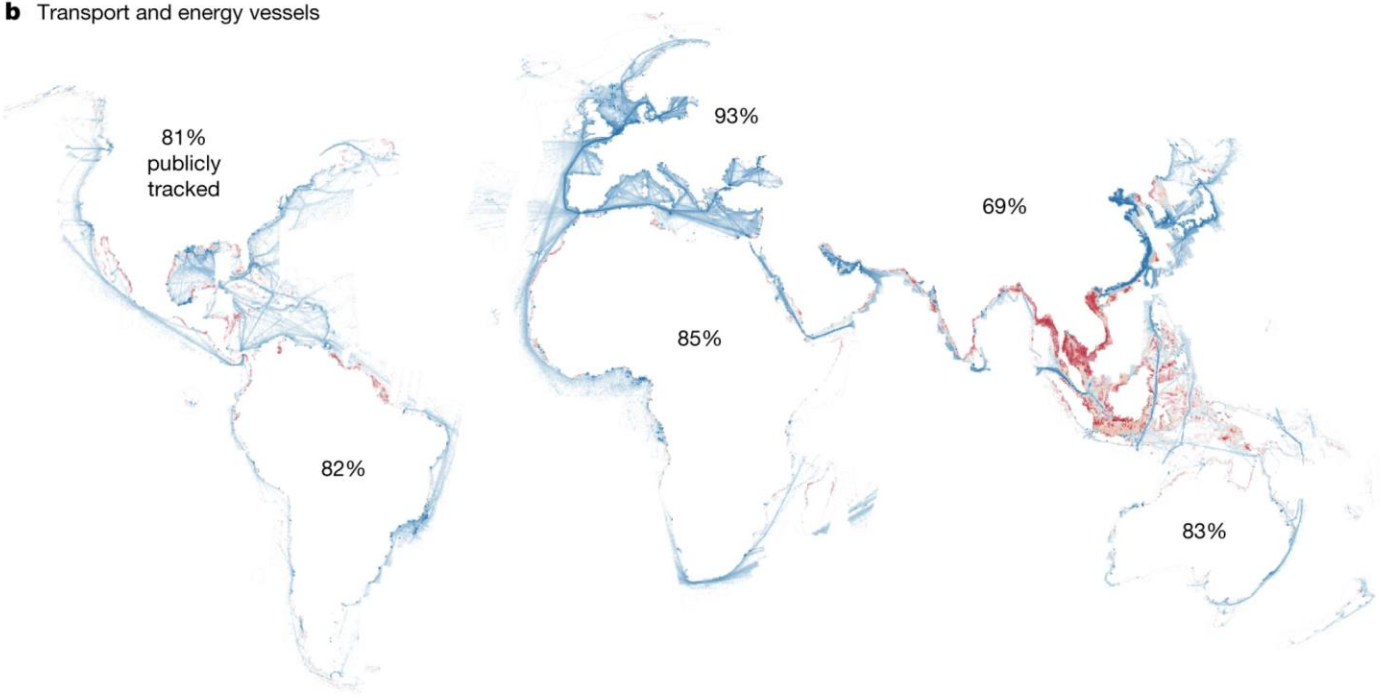
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Satellite Mapping Reveals Extensive Industrial Activity at Sea

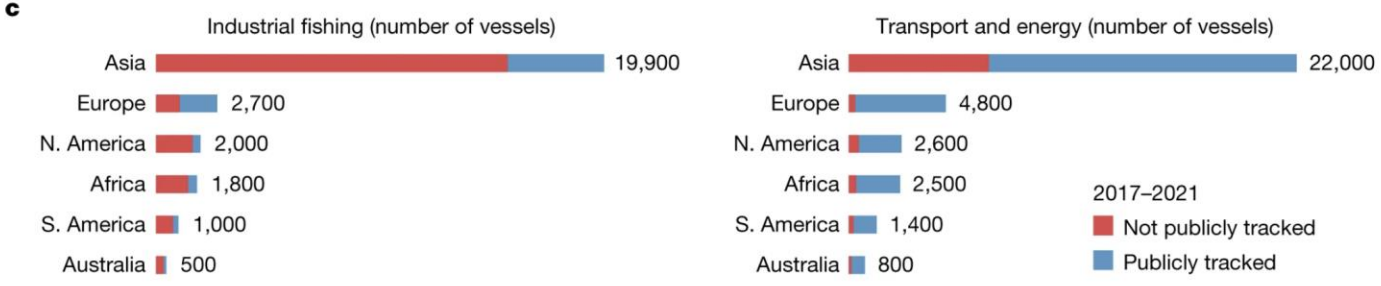
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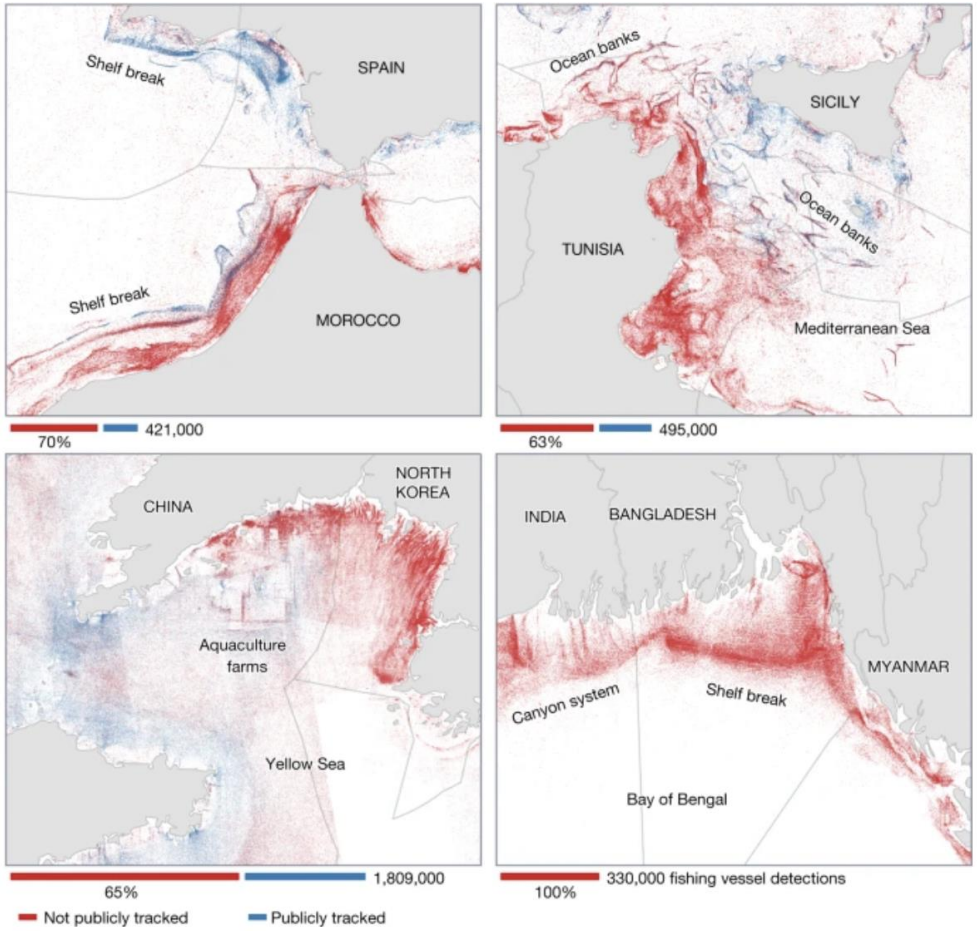

a Industrial fishing vessels
17%
publicly
tracked
61%
22%
22%
23%
25%
Fraction of vessels publicly tracked per km²
0%
25%
50%
75%
100%

b Transport and energy vessels



c




Shelf break
SPAIN
Shelf break
MOROCCO
70%
421,000
Ocean banks
SICILY
Ocean banks
TUNISIA
Mediterranean Sea
63%
495,000
NORTH KOREA
CHINA
Aquaculture farms
Yellow Sea
65%
1,809,000
Not publicly tracked
Publicly tracked
INDIA
BANGLADESH
MYANMAR
Canyon system
Shelf break
Bay of Bengal
100%
330,000 fishing vessel detections

Satellite Mapping: Key Findings

Source: Nature (2024)

Around 75% of industrial fishing is not publicly tracked, along with about 25% of other vessel activity

High-resolution mapping: reveals fishing patterns missing from public tracking systems

COVID-19 impact: fishing dropped sharply; transport & energy were relatively unaffected

Offshore footprint: wind turbines surpassed oil structures in 2021

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Sourcing Alternative Data
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Origins of Alternative Data

1

Individuals

Social media, reviews, search trends

App usage, receipts, mobile geolocation

2

Business Processes

Card & point-of-sale transactions

Job postings, pricing, supply chain

3

Sensors

Satellite & aerial imagery

Foot-traffic sensors, IoT, telematics

Key insight: most alternative data is a by-product of some other activity. It was often not originally designed for analysis

Four Ways to Source It

1

Collect it yourself

Web scraping, public APIs, your own panels. Full control and possible exclusivity, but engineering-heavy.

2

Buy from vendors

Placer.ai, Orbital Insight, Yodlee / Second Measure. Analysis-ready, but costly and shared with others.

3

Marketplaces & scouts

Neudata, Eagle Alpha, Datarade, Snowflake, BattleFin. Help you discover, compare & trial datasets.

4

Partnerships

Direct data-sharing deals with the owner. Example: our Applied Data Challenge with Thurro

From Raw Feeds to Signals

Raw Feeds

Unprocessed exhaust: raw transactions and imagery. You clean it all.

Cheapest · most edge

Processed Data

Cleaned, de-duplicated, mapped to companies. Most common offering.

Analysis-ready

Signals & Scores

Vendor's own indicators, e.g. predicted revenue surprise. Easiest to use.

Least differentiated

Trade-off: the more refined the product, the less potential edge it carries.

Next week, the Data Value Chain: Acquisition → Cleaning → Enrichment → Delivery

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Free & Open Sources to Start With

Government & Filings

data.gov · US Census

Bureau of Labor Statistics · SEC EDGAR

Satellite & Geospatial

Landsat (NASA/USGS) · Sentinel

Earth Engine · Planetary Computer

Web & Trends

Google Trends · Wikipedia pageviews

Common Crawl · OpenStreetMap

Community & City Data

Kaggle · Hugging Face datasets

City open-data: bikes, taxis, 311

Great starting points for your term project, with no procurement required.

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How to Vet a Dataset

- ✓ **Coverage & history:** enough companies / regions, and enough history to backtest?
- ✓ **Point-in-time integrity:** does it show what was knowable then, or was history backfilled?
- ✓ **Frequency & latency:** daily or monthly, and how long until you actually receive it?
- ✓ **Panel quality:** is the sample big, stable and representative, or does it drift?
- ✓ **Mappability:** can records tie to companies, tickers, or business segments?
- ✓ **Legal & compliance:** any PII exposure, GDPR / CCPA, scraping ToS or MNPI risk?
- ✓ **Cost vs. expected edge:** can you trial it first, and who else already has it?

Sourcing Pitfalls & Red Flags

! **Backfilled history**

great-looking performance rebuilt after the fact

! **Survivorship bias**

failed names or churned users quietly dropped

! **Panel drift**

the user base shifts, so old and new aren't comparable

! **Tiny samples**

a few thousand users ≠ a national trend

! **Crowded signals**

everyone buys it, so the edge is already priced in

! **Unclear provenance**

vendor can't explain source or consent

Rule of thumb: If results look unusually strong, verify point-in-time integrity, provenance, and out-of-sample performance before trusting them.

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Sourcing & Your Term Project

IDEA PITCH DUE

Sep 16

Data sourcing should be part of your pitch.

Before you pitch, ask yourself:

? Can I actually obtain this data? (free tier, academic access, feasible scraping)

? Is there enough history & coverage for meaningful analysis?

? Can I legally and ethically use it? (license, terms of service, privacy)

- ✓ Prefer data you can take end-to-end: acquisition → cleaning → insight
- ✓ Document provenance & licensing from day one
- ✓ Bring sourcing questions to TA office hours

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Out of Class Activity

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Brewing Success: Winning Strategies for Coffee Chains

Placer.ai Whitepaper

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Your Task: The Placer.ai Whitepaper

Read **Brewing Success: Winning Strategies for Coffee Chains** (Placer.ai)

Prepare to discuss next class (Week 3):

- **Standout strategy:** which brand's growth strategy stood out, and why?
- **Consumer behavior:** how do short visits, suburban families & commuters change outcomes?
- **Expansion trade-offs:** can a chain grow without cannibalizing existing stores?

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Podcast Spotlight for Next Week

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Podcast: All About Alternative Data

J.P. MORGAN · MAKING SENSE

Trading Insights: All About Alternative Data

23 minutes · May 2025

Mark Fleming-Williams, Head of Data Sourcing at CFM, in conversation with Eloise Goulder of J.P. Morgan.

Listen for three things

- How the sector grew up, and why COVID was the moment location and card data proved themselves.
- How a fund decides a dataset is worth paying for, which is the checklist from the vetting slide.
- What AI and large language models are doing to the industry you are about to walk into.

Listen free on the J.P. Morgan site, with a full transcript, or search Making Sense on Apple Podcasts, Spotify or YouTube.

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Thank You & Questions

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About Thurro

A structured-intelligence platform purpose-built for Indian capital markets.

790+	3B+	470K	25–35M
datasets in the catalogue	structured observations	research documents	new data points / day

Products built on a proprietary DataLakehouse:

AltData	Thurro Answers	NowCast
Curated, visualised repository of alternative data	Conversational AI that turns a question into a decision-ready answer in seconds	Predictive tracking, ~70% accuracy up to 90 days ahead

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The Data: 34 Datasets, 6 Verticals

Consumer & B2C

VAHAN, UPI Plus, EcomTrack, QCOM Watch, DineTrack, StoreScope, FareWatch

Sector & supply-side

PharmaChain, PowerScope, TelcoLens, HealthGrid, RERA Watch, WorkPulse

Energy, metals & industrial

EnergyMap, MetalsLens, CapexLeads, LogiTrack, B2BTrade

Financial & capital markets

FundFlow, CreditPulse, FlowDesk, InsureMetrics, CreditForensic, EventDeck

Macro, fixed income & trade

TaxPulse, DebtScope, TradeLens

Agri, rural & documents

AgriPulse, Thurro1Reports (research corpus)

Getting Access to the Data

API: ai.thurro.com/api-docs, institutional-grade REST endpoints

MCP server: Thurro1DB / Thurro1MF / Thurro1Reports, built for AI and LLM workflows

How You'll Be Evaluated

Sourcing rigor

Right datasets; provenance, coverage & licensing understood

Signal quality

Data actually supports the claim; noise handled honestly

Analysis depth

Clean, combine & interrogate; sound, reproducible method

Insight / so-what

A clear, decision-relevant conclusion, not just charts

Communication

Tells the story crisply; well-structured & visual

Support & Timeline

Akhilesh, Co-founder at Thurro
Guest next class · akhilesh@thurro.com

TA office hours: bring your sourcing & data questions

Scoping help: Thurro can help scope the right datasets

Key date

Sep 16

Idea pitch due

Data sourcing should be part of your pitch.